

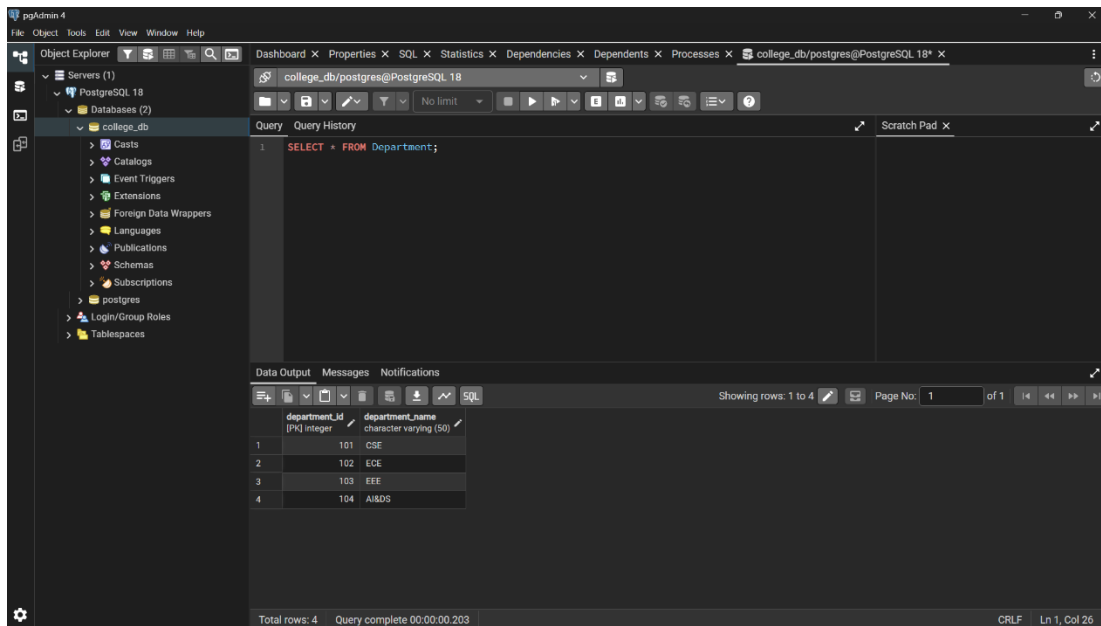
DSA0503 – Query Processing for Data Science

Activity using PostgreSQL

Divyesh S P (192424147)

1. Department Table (Parent Table) with attributes Department_ID

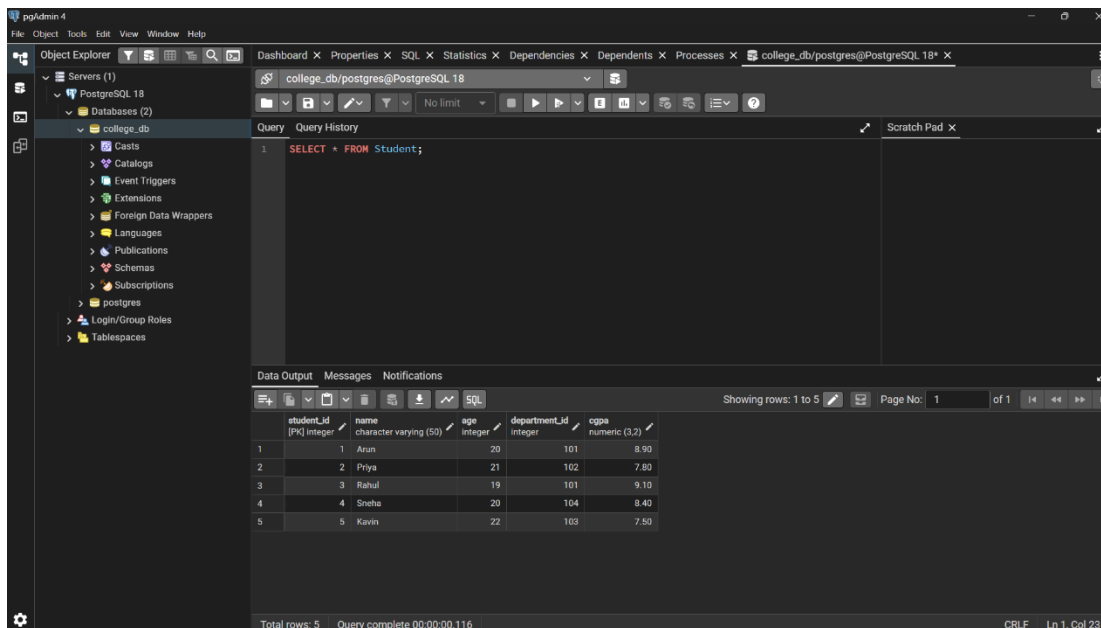
INTEGER PRIMARY KEY, Department_Name TEXT



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure with 'college_db' selected. The central pane shows the SQL query editor with the query: `SELECT * FROM Department;`. The bottom pane displays the data output for the 'Department' table.

department_id	department_name
101	CSE
102	ECE
103	EEE
104	Ai&DS

2. The Student Table (Child Table)



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure with 'college_db' selected. The central pane shows the SQL query editor with the query: `SELECT * FROM Student;`. The bottom pane displays the data output for the 'Student' table.

student_id	name	age	department_id	cgpa
1	Anu	20	101	8.90
2	Priya	21	102	7.80
3	Rahul	19	101	9.10
4	Sneha	20	104	8.40
5	Kavin	22	103	7.50

3. Displayed students from CSE department.

The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'college_db/postgres@PostgreSQL 18*' database. The query is as follows:

```
1 SELECT Student.Student_ID,  
2       Student.Name,  
3       Student.Age,  
4       Student.CGPA,  
5       Department.Department_Name  
6 FROM Student  
7 JOIN Department  
8 ON Student.Department_ID = Department.Department_ID  
9 WHERE Department.Department_Name = 'CSE';
```

The Data Output pane displays the following table:

student_id	name	age	cgpa	department_name
1	Arun	20	8.90	CSE
2	Rahul	19	9.10	CSE

The status bar at the bottom indicates: Total rows: 2, Query complete 00:00:00.110, and a success message: Successfully run. Total query runtime: 110 msec. 2 rows affected.

4. Counted students in each department.

The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'college_db/postgres@PostgreSQL 18*' database. The query is as follows:

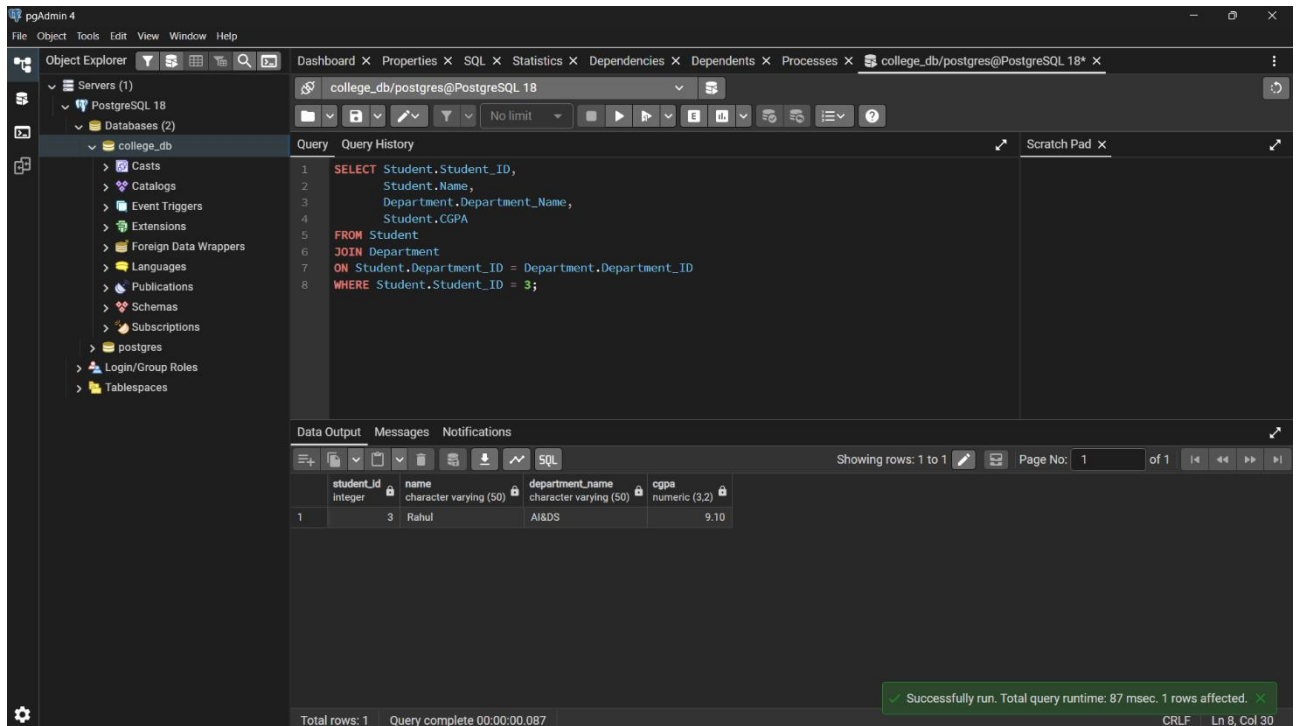
```
1 SELECT Department.Department_Name,  
2       COUNT(Student.Student_ID) AS Total_Students  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 GROUP BY Department.Department_Name;
```

The Data Output pane displays the following table:

department_name	total_students
CSE	2
ECE	1
AI&DS	1
EEE	1

The status bar at the bottom indicates: Total rows: 4, Query complete 00:00:00.108, and a success message: Successfully run. Total query runtime: 108 msec. 4 rows affected.

5. Updated the foreign key.



The screenshot shows the pgAdmin 4 interface with a SQL query being executed. The query updates the foreign key for a student in the 'college_db' database. The query is as follows:

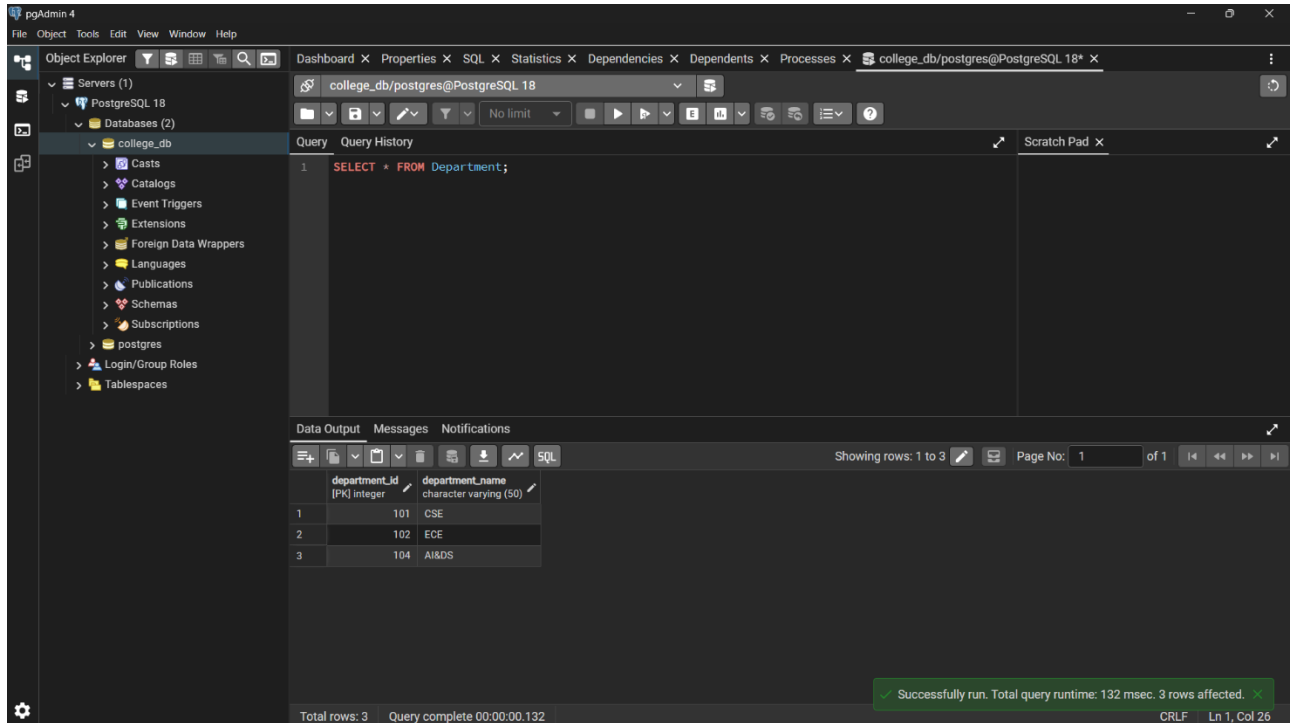
```
1 SELECT Student.Student_ID,  
2       Student.Name,  
3       Department.Department_Name,  
4       Student.CGPA  
5 FROM Student  
6 JOIN Department  
7 ON Student.Department_ID = Department.Department_ID  
8 WHERE Student.Student_ID = 3;
```

The result is displayed in the Data Output tab, showing one row of data:

student_id	name	department_name	cgpa
3	Rahul	AI&DS	9.10

A status message at the bottom indicates: "Successfully run. Total query runtime: 87 msec. 1 rows affected."

6. Deleted a department



The screenshot shows the pgAdmin 4 interface with a SQL query being executed. The query deletes a department from the 'college_db' database. The query is as follows:

```
1 SELECT * FROM Department;
```

The result is displayed in the Data Output tab, showing three rows of data:

department_id	department_name
101	CSE
102	ECE
104	AI&DS

A status message at the bottom indicates: "Successfully run. Total query runtime: 132 msec. 3 rows affected."

7. Displayed each student's name along with the corresponding department name using an SQL JOIN.

The screenshot shows the pgAdmin 4 interface with the 'college_db/postgres@PostgreSQL 18*' connection selected. The 'Query' tab is active, displaying the following SQL query:

```
1 SELECT Student.Name,  
2       Department.Department_Name  
3 FROM Student  
4 INNER JOIN Department  
5 ON Student.Department_ID = Department.Department_ID;
```

The 'Data Output' tab shows the results of the query, displaying 5 rows. The columns are 'name' (character varying (50)) and 'department_name' (character varying (50)).

name	department_name
Arun	CSE
Priya	ECE
Sneha	AI&DS
Rahul	AI&DS
Kavin	CSE

The status bar at the bottom indicates: 'Total rows: 5 Query complete 00:00:00.144'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 144 msec. 5 rows affected.'

8. Displayed the number of students in each department.

The screenshot shows the pgAdmin 4 interface with the 'college_db/postgres@PostgreSQL 18*' connection selected. The 'Query' tab is active, displaying the following SQL query:

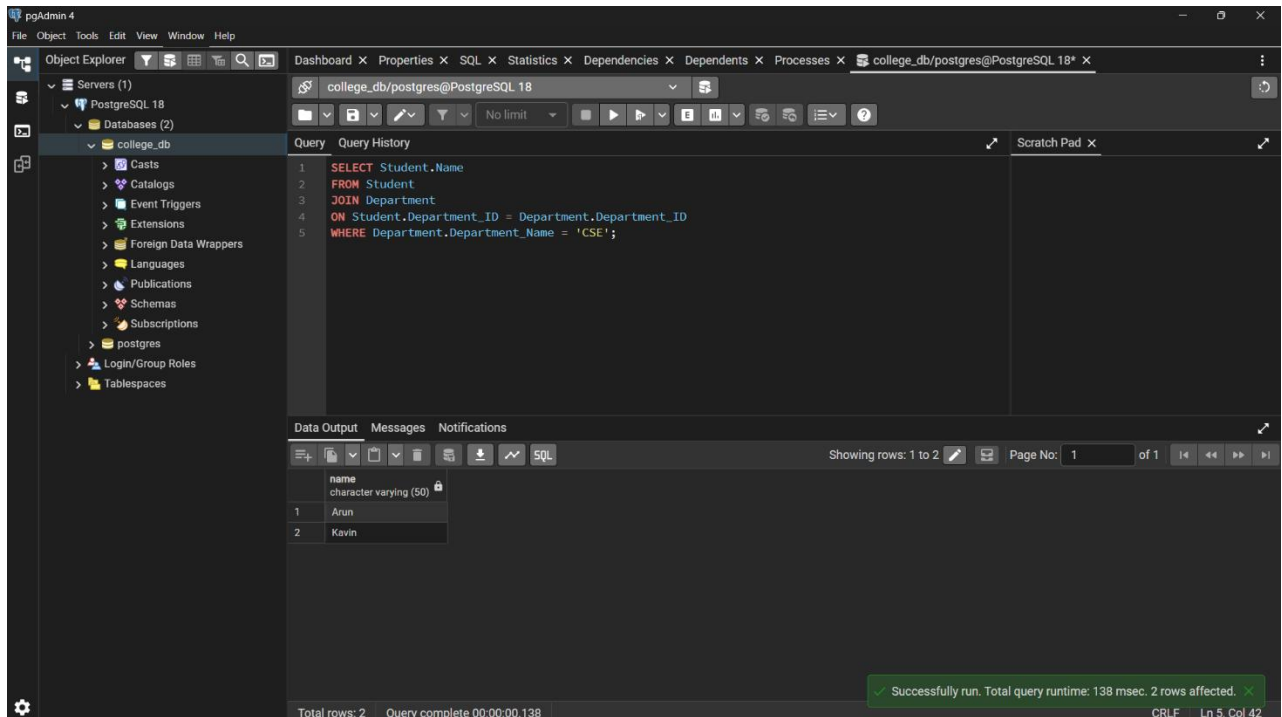
```
1 SELECT Department.Department_Name,  
2       COUNT(Student.Student_ID) AS Number_of_Students  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 GROUP BY Department.Department_Name;
```

The 'Data Output' tab shows the results of the query, displaying 3 rows. The columns are 'department_name' (character varying (50)) and 'number_of_students' (bigint).

department_name	number_of_students
CSE	2
ECE	1
AI&DS	2

The status bar at the bottom indicates: 'Total rows: 3 Query complete 00:00:00.120'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 120 msec. 3 rows affected.'

9. Displayed the names of students belonging to the Computer Science department.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database. The main pane shows a SQL query executed against the 'college_db/postgres@PostgreSQL 18*' connection. The query is as follows:

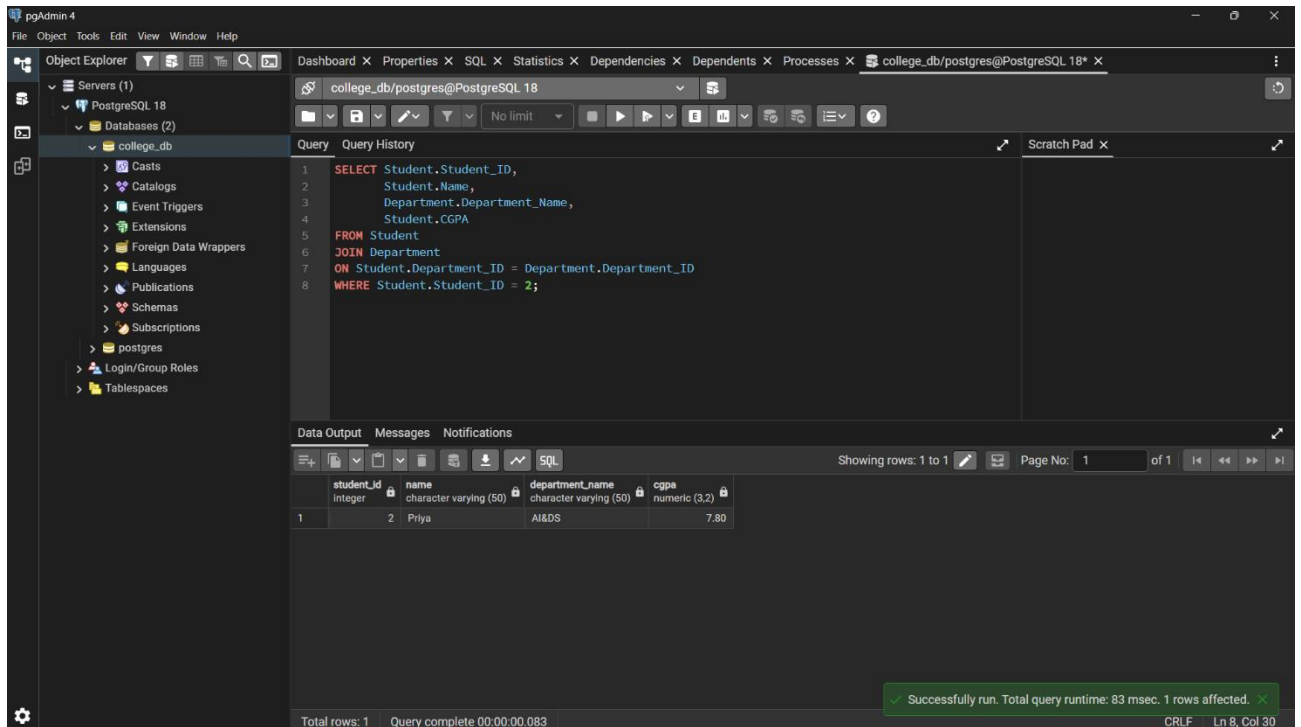
```
1 SELECT Student.Name
2 FROM Student
3 JOIN Department
4 ON Student.Department_ID = Department.Department_ID
5 WHERE Department.Department_Name = 'CSE';
```

The 'Data Output' tab at the bottom shows the results of the query, displaying two rows of student names:

name
Arun
Kavin

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:00.138'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 138 msec. 2 rows affected.'

10. Update the department of a student by changing the foreign key value and display the updated record.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database. The main pane shows a SQL query executed against the 'college_db/postgres@PostgreSQL 18*' connection. The query is as follows:

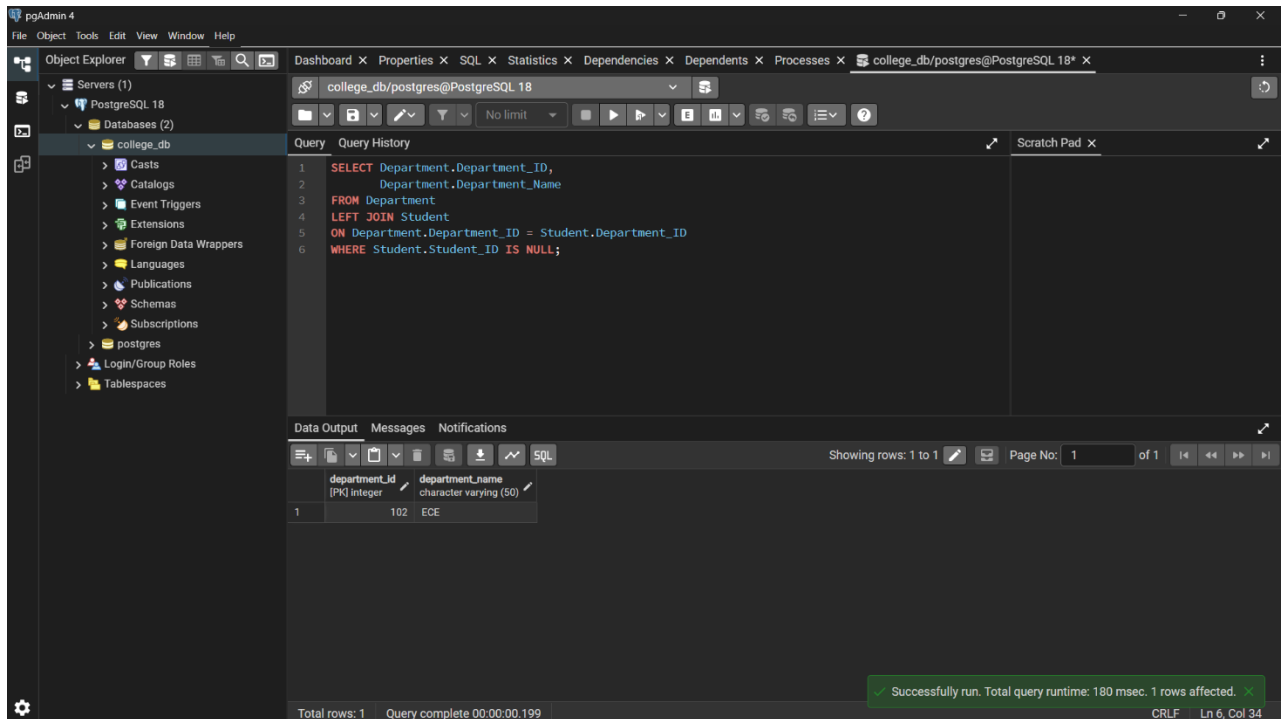
```
1 SELECT Student.Student_ID,
2       Student.Name,
3       Department.Department_Name,
4       Student.CGPA
5 FROM Student
6 JOIN Department
7 ON Student.Department_ID = Department.Department_ID
8 WHERE Student.Student_ID = 2;
```

The 'Data Output' tab at the bottom shows the results of the query, displaying one row of student information:

student_id	name	department_name	cgpa
2	Priya	AI&DS	7.80

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.083'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 83 msec. 1 rows affected.'

11. Display departments that currently have no students assigned using an appropriate SQL JOIN.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database. The main pane shows a SQL query that uses a LEFT JOIN to find departments with no students assigned. The query is as follows:

```
1 SELECT Department.Department_ID,  
2      Department.Department_Name  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 WHERE Student.Student_ID IS NULL;
```

The 'Data Output' pane shows the results of the query, which is a single row representing the department with ID 102 and name 'ECE'.

department_id	department_name
102	ECE

The status bar at the bottom indicates that the query was successfully run, with a total query runtime of 180 msec and 1 row affected.